

CSE 321 Database Systems

Lab Session #1 (2026 Version)

ER Modelling and Reflective Review Report

Overview

In this lab session, you will construct an ER diagram from a given specification during class. The purpose of this activity is not only to practise conceptual database design, but also to improve your ability to identify modelling errors, clarify assumptions, and explain design decisions in a precise and logical manner.

This lab is designed as an **individual** exercise. You are expected to complete the ER modelling task during the lab session and later prepare a short reflective report based on your own submitted work.

Deadline

The submission deadline is **April 8, 23:59**. Please submit your assignment in advance to avoid any last-minute issues. Late submissions will receive a score of 0.

In-Class Activity

During Lab Session #1, a specification will be provided in class. Based on that specification, you must draw an ER diagram using **Chen's notation**.

You should aim to produce a diagram that is as complete, clear, and logically consistent as possible based on the given specification. If some parts are ambiguous, you can put your assumption in the bottom of the submitted ER-diagram. Attending the in-class lab session activity is mandatory.

After the Lab Session

After the lab session, I will collect all submissions and scan them. The scanned version of your own submission will then be distributed back to you individually.

At that point, you must carefully review your original ER diagram again by revisiting the specification. If necessary, you may also use LLM for this task. The purpose of this step is **not** simply to correct the answer, but to analyse your own modelling process in a critical and transparent way.

You should identify:

- what you modelled correctly,
- what you modelled incorrectly,
- what you misunderstood in the specification,

- what assumptions you made,
- whether those assumptions were reasonable,
- how and why you revised the diagram.

Report Submission

You must prepare an individual report in **English** using **Overleaf**. I will upload the report template separately, and you must use that template for your submission.

The report must be written in a clear and organised manner, with a maximum length of **10 pages**.

Your report must include all of the following components:

- the **original ER diagram** that you submitted during the lab session,
- the **revised ER diagram**,
- a clear explanation of **each modification**,
- a discussion of any misunderstanding, omission, ambiguity, or incorrect assumption in your original work,
- a concise justification for the final revised design.

A good report is expected to do more than present two diagrams. It should clearly explain the difference between the original and revised versions, and it should demonstrate that you understand **why** the revision was necessary.

Expected Report Structure

You may organise your report in any reasonable way, but the following structure is strongly recommended:

1. **Introduction:** Briefly explain the purpose of the report.
2. **Review of the Original Design:** First, present the exact ER diagram you submitted during the lab session by digitally recreating it and inserting it using the **figure*** environment. Then, explain your reasoning behind this initial design. Focus on detailing the modeling logic you applied and any assumptions you made during the in-class activity.
3. **Revision Details:** First, present your fully corrected ER diagram by digitally creating it and inserting it using the **figure*** environment. Then, explain each revision clearly and specifically. For every modification, insert cropped images of the original and revised parts side-by-side for direct comparison, followed by your explanation of why the change was necessary. Specifically, you need to explain (1) what was wrong in the original modelling; (2) why it was wrong from a conceptual modelling perspective; and (3) how the revised design resolves the issue. Please refer to the example format for details.
4. **Conclusion:** Summarise what you learned from the exercise.

Evaluation

The grading criterion for this lab is intentionally simple.

- If the report is written clearly and demonstrates careful reflection and explanation, it will receive **4/4**.
- If the report is incomplete, unclear, or lacks sufficient explanation, it may receive **3/4** or lower.
- If you do not attend the lab activity session, or if you submit after the deadline, the score will be **0**, regardless of the reason.

In other words, this lab is primarily evaluated based on the **clarity, completeness, and sincerity of your reflective report**, not solely on whether your original ER diagram was perfect.

Final Remark

This lab is intended to help you develop an important skill in database design: not only drawing an ER diagram, but also reviewing your own design critically, identifying where your reasoning failed, and explaining how to improve it in a structured way.

Using LLM tools to generate the entire report without genuine reflection will be considered inappropriate and may result in a reduced score.

A correct diagram is valuable, but a clear and honest explanation of your modelling process is equally important.